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Revision History

Rev	Description	Date	Approved/Designer
X	Initial Schematic	OCT-2024	Efren D/ Antonio Q
X4	A070 Release	11/12/24	Efren D/ Antonio Q
A	BOM Optimization TPs removed (OSDA)	11/28/24	Efren D/ Antonio Q
	R17 to zero ohm resistor SW1 removed, VBUS_IN to jack (J10) connector FXLS8964AF added	12/11/24	Efren D/ Antonio Q
	Temp Sensor (U18) added	12/13/24	Efren D/ Antonio Q
	A085 Release	12/18/24	Efren D/ Antonio Q
A1	Y2 replacement	01/14/25	Efren D/ Antonio Q
AX1	Test points cleaning Added ethernet strap-in resistors Change on VBUS domain levels on CAN/LIN	05/09/25	Efren Diaz
	Rebranding name	05/20/25	
	C161, C162, R731 added	05/27/25	Efren D/ Antonio Q
	SJ6, R732, C163 added	06/06/25	
	Replace SJ6 for JP1 (3x1 header)		
B	Release	07/08/25	Efren D/ Antonio Q
B1	R190 to DNP, R733 populated	10/01/25	Efren D/ Antonio Q

S32K344MINI-EVB

" TLALOC "

Customer Evaluation Board

C A U T I O N :

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Signals (ports) have not been routed via busses as this makes it harder to determine where each signal goes.

User notes are given throughtout the schematics.

Specific PCB LAYOUT notes are detailed in ITALICS

Notes:

- All components and board processes are to be ROHS compliant
- All connectors and headers are denoted Jx/JPx and are 2.54mm pitch unless otherwise stated
- All jumpers are denoted Jx. Jumpers are 2mm pitch
- Jumper default positions are shown in the schematics. For 3 way jumpers, default is always posn 1-2.
  2 Pin jumpers generally have the "source" on pin 1.
- All switches are denoted SWx
- All test points are denoted TPx

2 Different test points used in design:

TPx - Through Hole Pad small

TPHx - Through Hole Pad Large (for standard 0.1" header). Also used on IO Matrix (IOMx)



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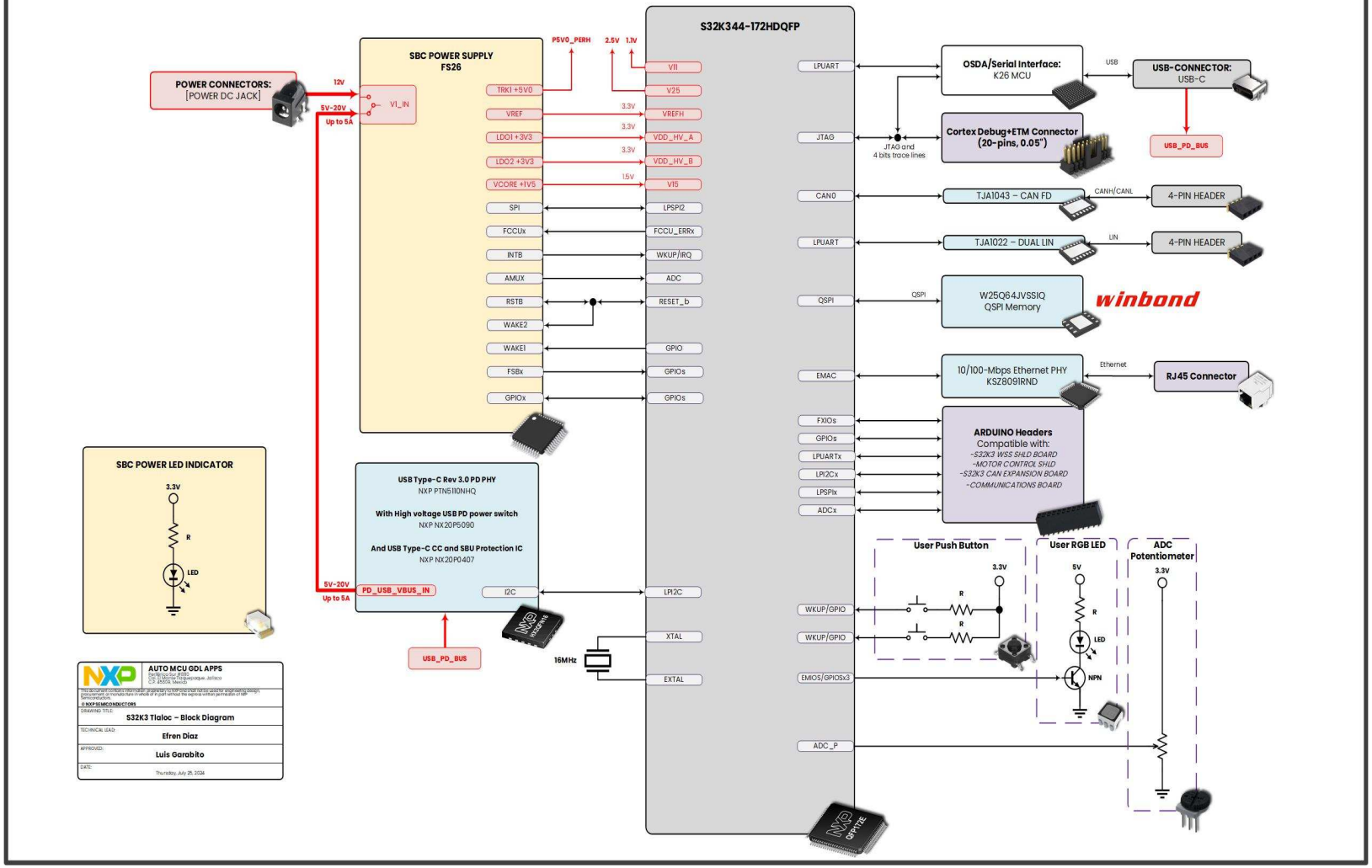
Designer: Antonio Q. / Efren D.	Drawing Title: S32K344MINI-EVB	
Drawn by: Antonio Quiroz	Page Title: TITLE PAGE	
Approved: Efren Diaz	Size: C	Document Number: SCH-94921 PDF: SPF-94921 — Rev: B1
	Date: Wednesday, October 01, 2025	Sheet 1 of 14

S32K3 Tlaloc

S32K344MINI-EVB

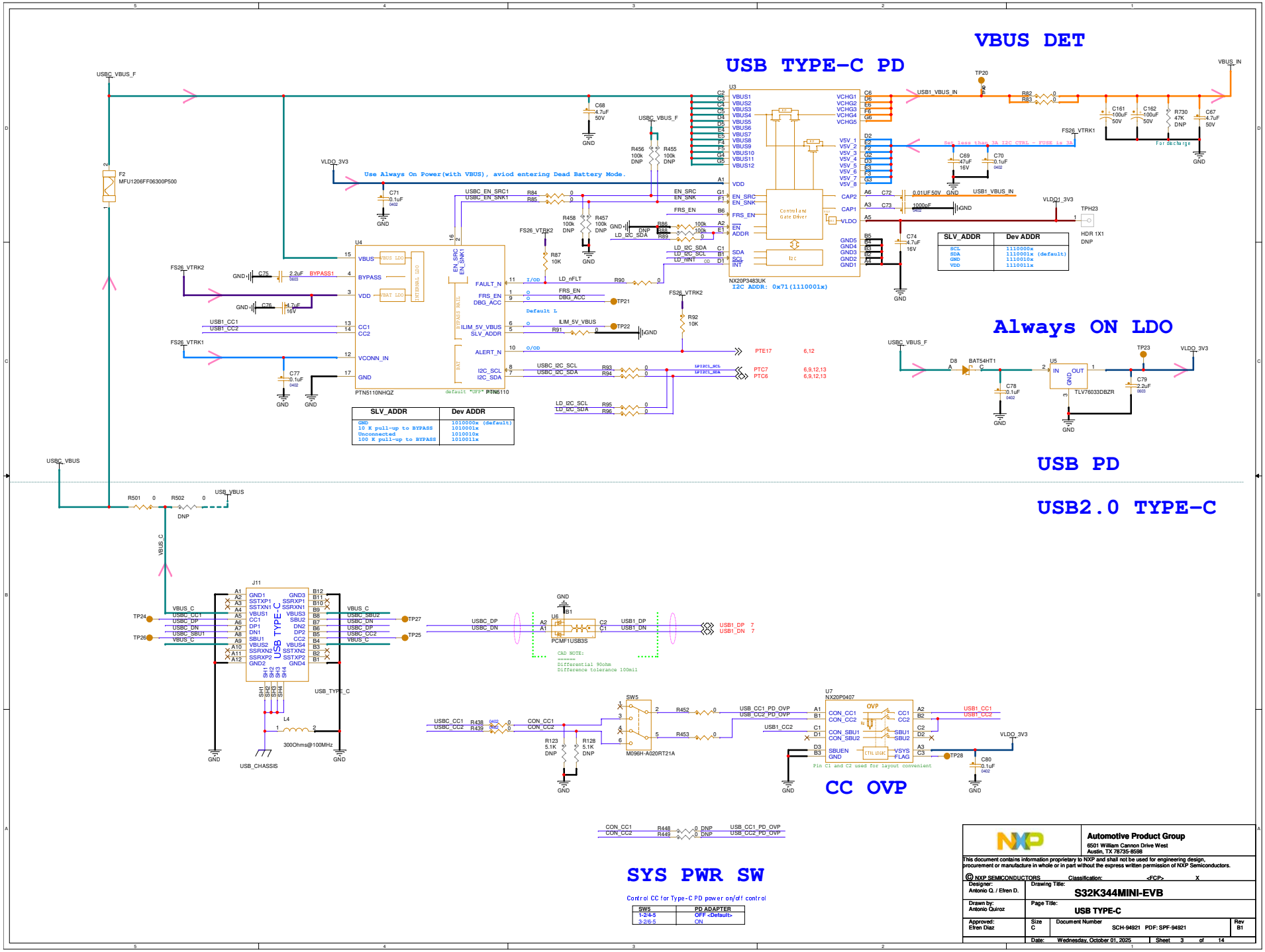
S32K344 HDQFP172 Tlaloc Evaluation Board

05/27/2024 Rev B.



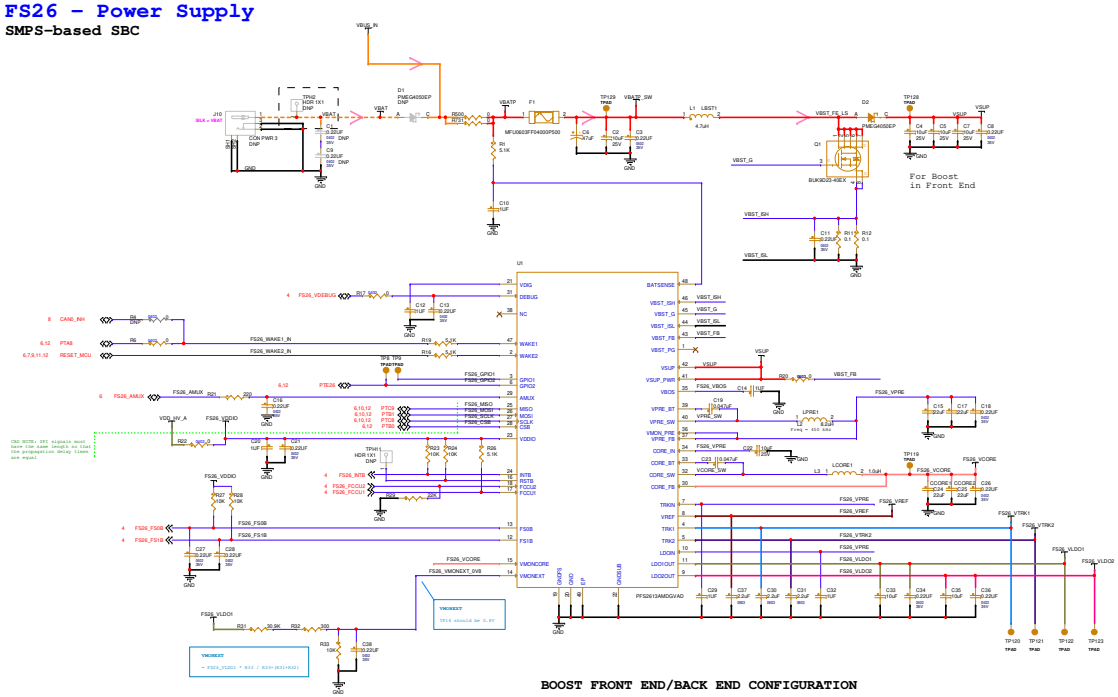
AUTO MCU GDL APPS	
DRAWING TITLE:	S32K3 Tlaloc – Block Diagram
TECHNICAL LEAD:	Efren Diaz
APPROVED:	Luis Garabito
DATE:	Thursday, July 25, 2024

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Drawn by: Antonio Quiroz	Page Title:	NOTES & BLOCK DIAGRAM	
Approved: Efren Diaz	Size: C	Document Number: SCH-94921 PDF: SPF-94921	Rev: B1
	Date: Wednesday, October 01, 2025	Sheet 2 of 14	

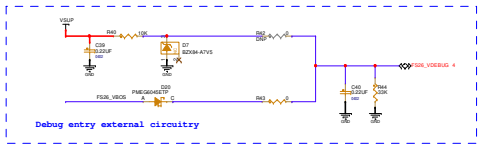

VBUS DET
USB TYPE-C PD
Always ON LDO
USB PD
USB2.0 TYPE-C
CC OVP
SYS PWR SW
USBC_VBUS_F
VBUS_IN
USB1_VBUS_IN
VLDO_3V3
F2
MFU1206FF06300P500
Use Always On Power(with VBUS), avoid entering Dead Battery Mode.
U3
NX20P3483UK
I2C ADDR: 0x71(1110001x)
SLV_ADDR
Dev ADDR
1110000x
1110001x (default)
1110010x
1110011x
U4
PTN5110NHQZ
default "UFP" PTN5110
SLV_ADDR
Dev ADDR
1010000x (default)
1010001x
1010010x
1010011x
TP20
TP21
TP22
TP23
TP24
TP25
TP26
TP27
TP28
TPH23
HDR 1X1
DNP
TLV76033DBZR
PCMF1USB3S
M096H-A020RT21A
NX20P0407
J11
USB TYPE-C
USB_TYPE_C
USB_CHASSIS
300Ohms@100MHz
SW5
USB_VBUS
VBUS_C
FS26_VTRK1
FS26_VTRK2
PTE17
PTC7
PTC6
6,12
6,9,12,13
6,9,12,13
USB1_DP
USB1_DN
USB1_CC1
USB1_CC2
USBC_DP
USBC_DN
USBC_CC1
USBC_CC2
USBC_SBU1
USBC_SBU2
CON_CC1
CON_CC2
USB_CC1_PD_OVP
USB_CC2_PD_OVP
Control and Gate Driver
OVP
CTRL LOGIC
Differential 90ohm
Difference tolerance 100mil
Pin C1 and C2 used for layout convenient
Control CC for Type-C PD power on/off control
SW5
PD ADAPTER
1-2/4-5
OFF <Default>
3-2/6-5
ON
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Classification: <FCP> X
Designer: Antonio Q. / Efren D.
Drawing Title: S32K344MINI-EVB
Drawn by: Antonio Quiroz
Page Title: USB TYPE-C
Approved: Efren Diaz
Size C
Document Number SCH-94921 PDF: SPF-94921
Rev B1
Date: Wednesday, October 01, 2025
Sheet 3 of 14

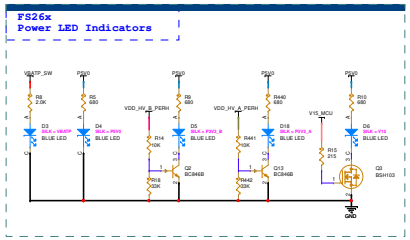
FS26 – Power Supply

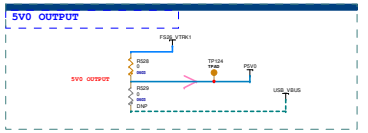
SMPS-based SBC

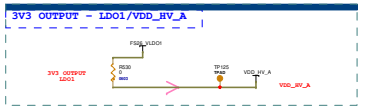


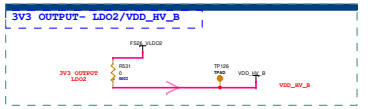
BOOST FRONT END/BACK END CONFIGURATION

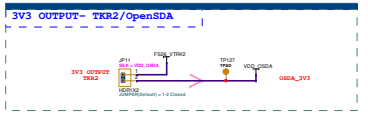












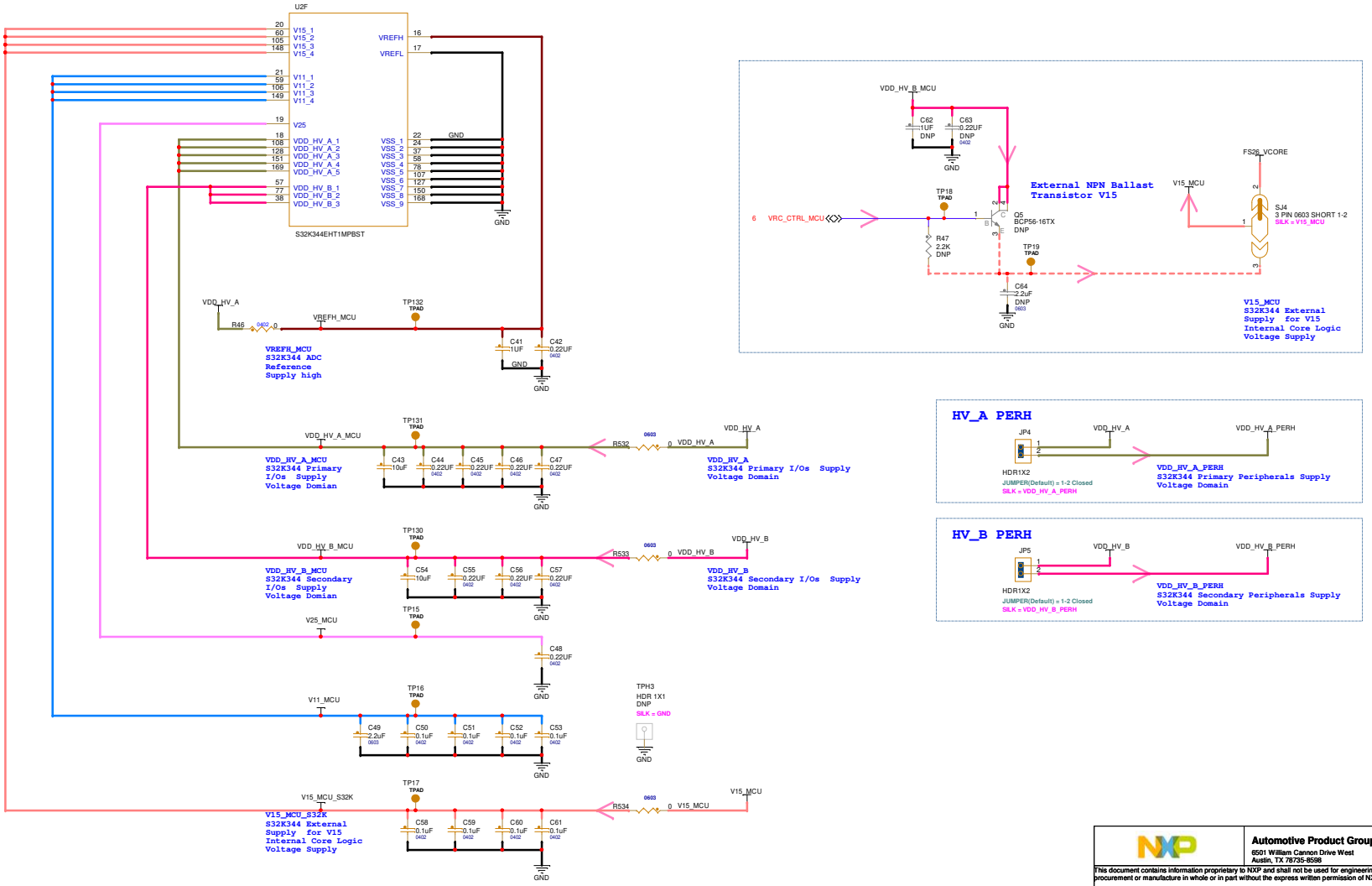
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Designer: Antonio G. / Efren D.	Drawing Title:	S32K344MINI-EVB	
Drawn by: Antonio Cuevz	Page Title:	FS26x SBC Power Supply	
Approved: Efren Diaz	Size: D	Document Number: SCH-94921 PDF:SPF-94921	Rev: B1
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S32K3XX Microcontroller – 172HDQFP

Power Supply Pins



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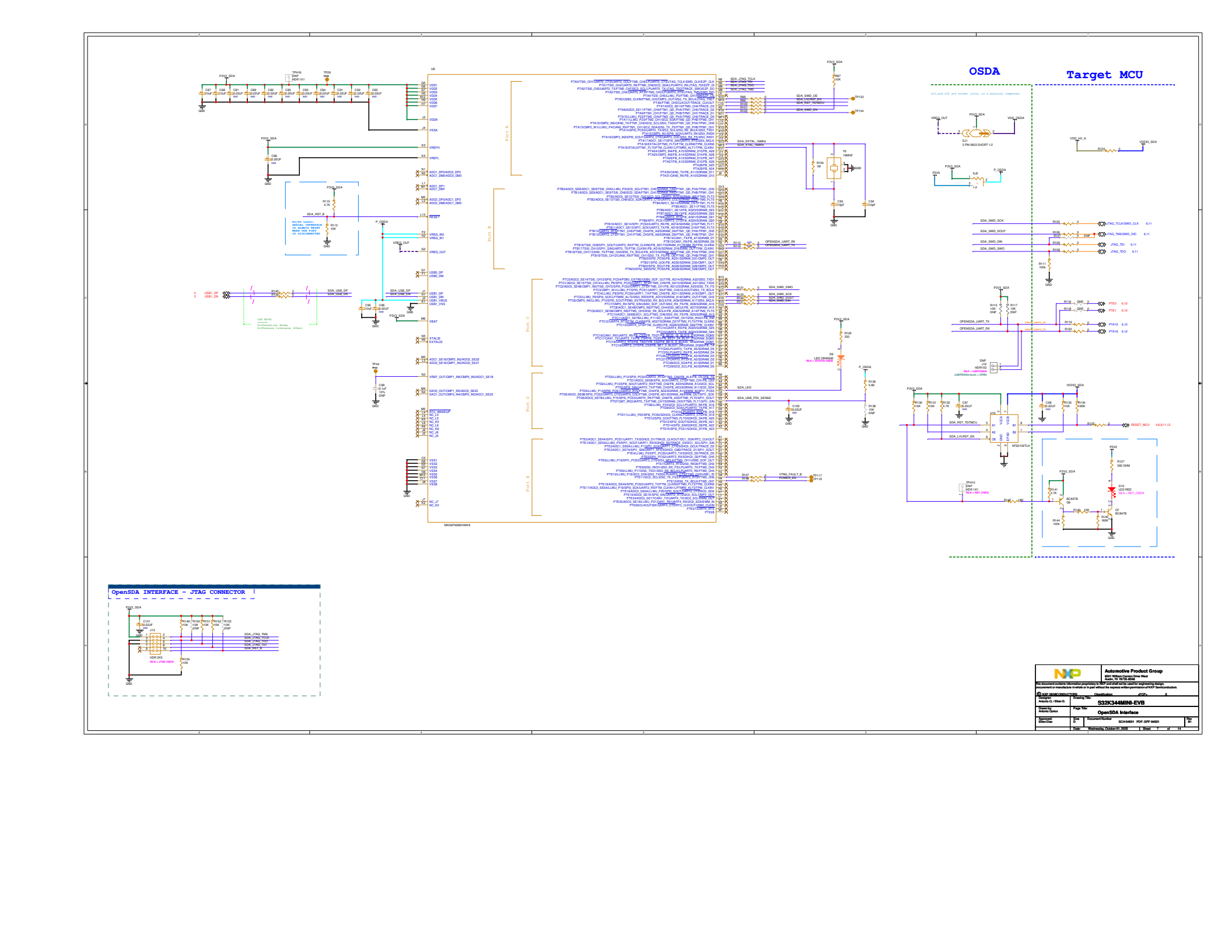
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Classification:	<FCP>	X	
Designer: Antonio Q. / Efren D.	Drawing Title: S32K344MINI-EVB		
Drawn by: Antonio Quiroz	Page Title: S32K3XX - MCU POWER Supply Pins		
Approved: Efren Diaz	Size: C	Document Number: SCH-94921 PDF: SPF-94921	Rev: B1
	Date: Wednesday, October 01, 2025	Sheet 5 of 14	

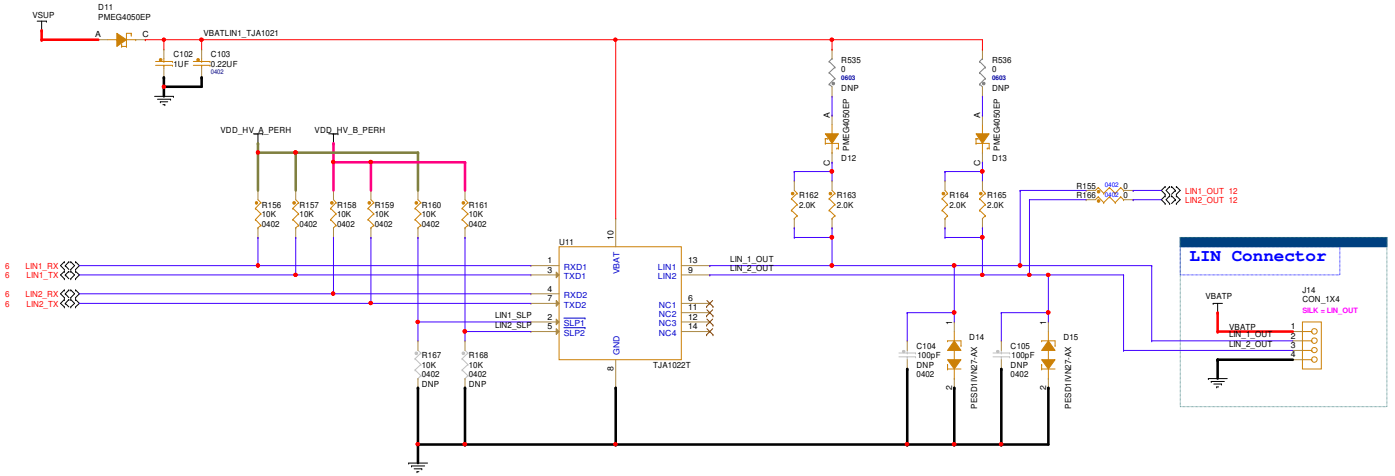
S32K3X4 Microcontroller – 172HDQFP

Ports A/B/C/D/E

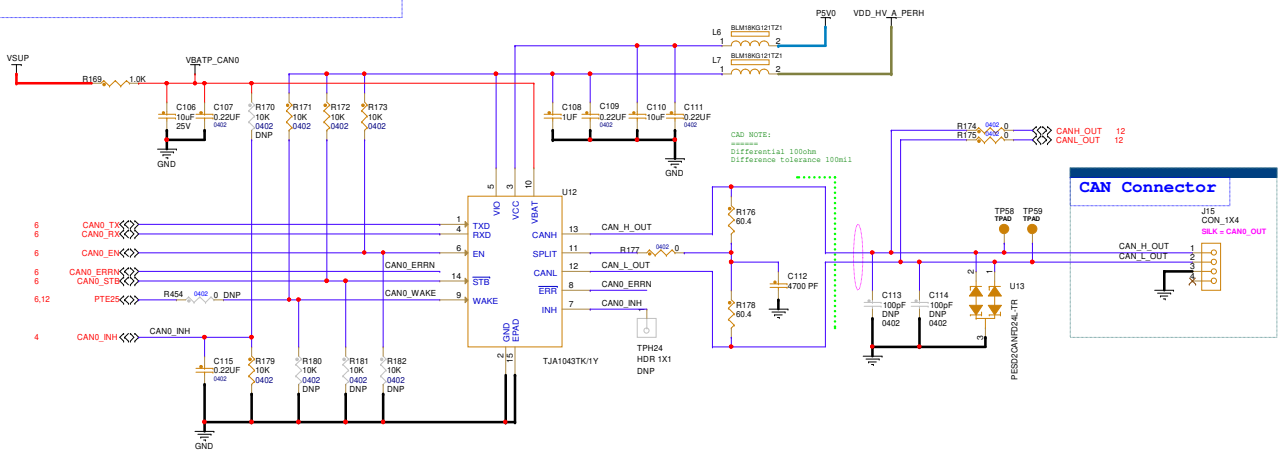



OSDA
Target MCU
OpenSDA INTERFACE - JTAG CONNECTOR
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S32K344MINI-EVB
OpenSDA Interface
Designer: Antonio G. / Efren D.
Drawn by: Antonio Cuevz
Approved: Efren Diaz
Size: D
Document Number: SCH-94921
Rev: B1
Date: Wednesday, October 01, 2025
Sheet 7 of 14

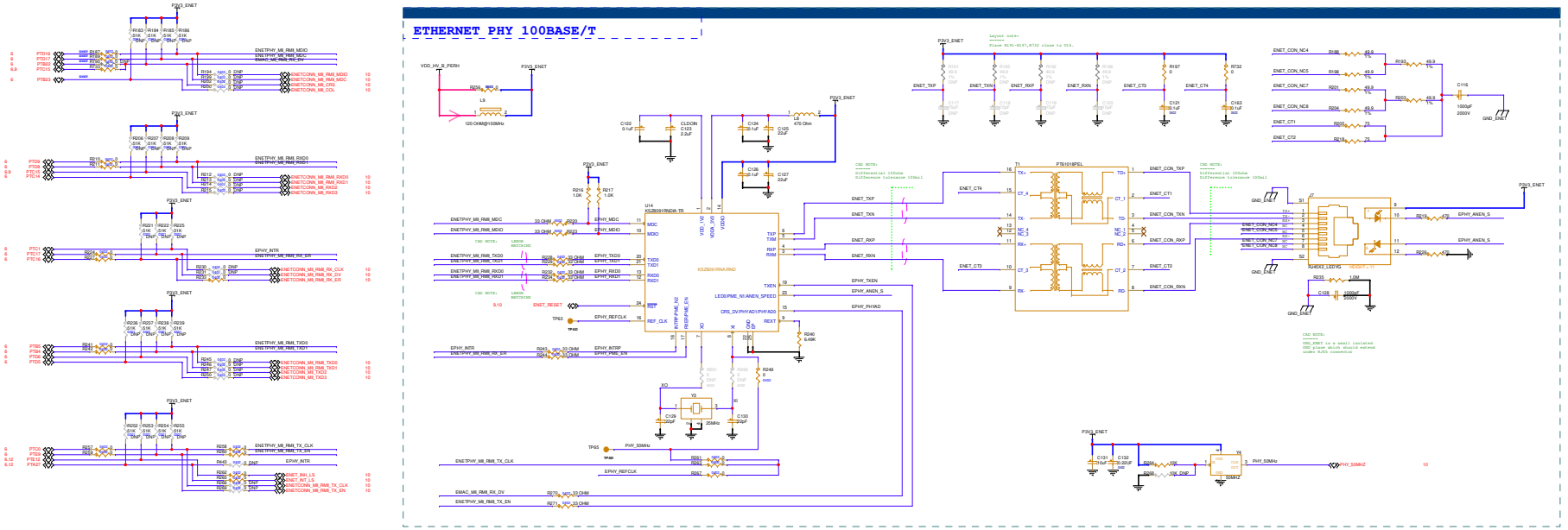
LIN Physical Layer
TJA1022T: Quad LIN 2.2A/SAE J2602 transceiver



TJA1043 HS-CAN Transceiver



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Designer: Antonio Q. / Efren D.	Drawing Title: **S32K344MINI-EVB**	
Drawn by: Antonio Quiroz	Page Title: **CAN/LIN PHY**	
Approved: Efren Diaz	Size C / Document Number SCH-94921 PDF: SPF-94921	Rev B1
	Date: Wednesday, October 01, 2025	Sheet 8 of 14

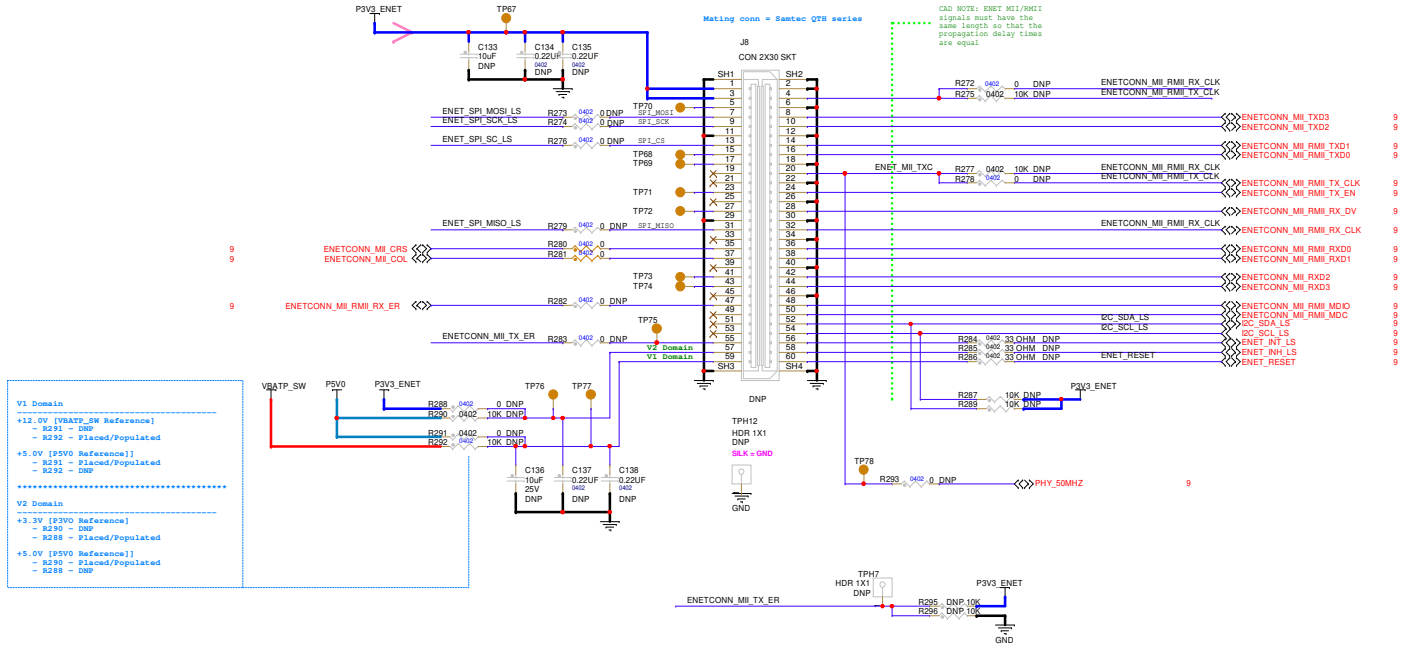


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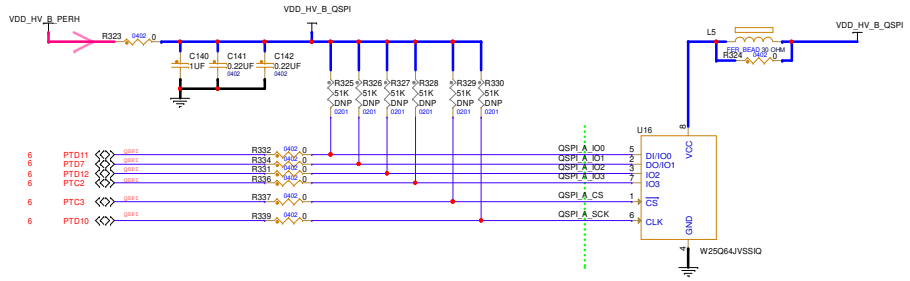
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Designer: Antonio G. / Efren D.	Drawing Title: S32K344MINI-EVB
Drawn by: Antonio Cuevz	Page Title: Ethernet Connector
Approved: Efren Diaz	Size: D; Document Number: SCH-94921 PDF:SPF-94921; Rev: B1
	Date: Wednesday, October 01, 2025; Sheet 8 of 14

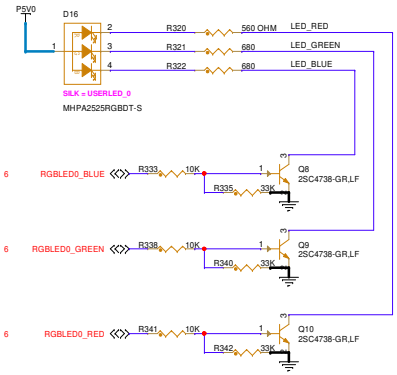


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Page Title:	ETHERNET CONNECTOR
Size: C	Document Number: SCH-94921 PDF: SPF-94921
Rev: B1	
Date:	Wednesday, October 01, 2025
Sheet	10 of 14

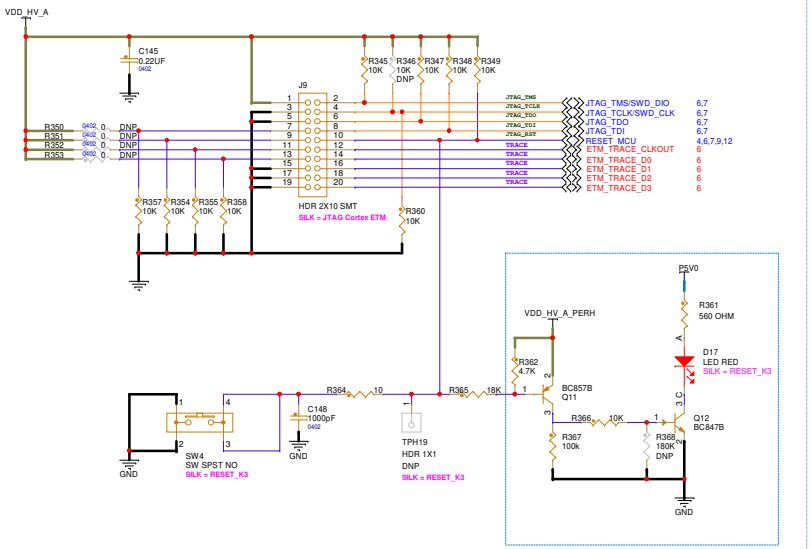
QSPI



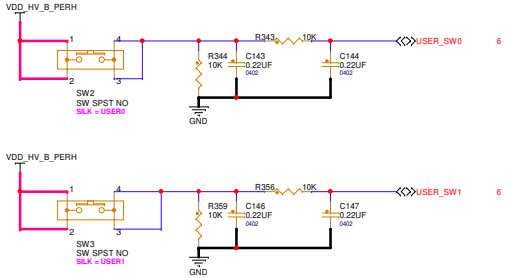
User RGB LED



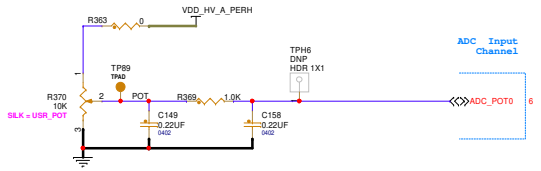
20-pin Cortex Debug D ETM connector



User Push Button - Active to HIGH/LOW



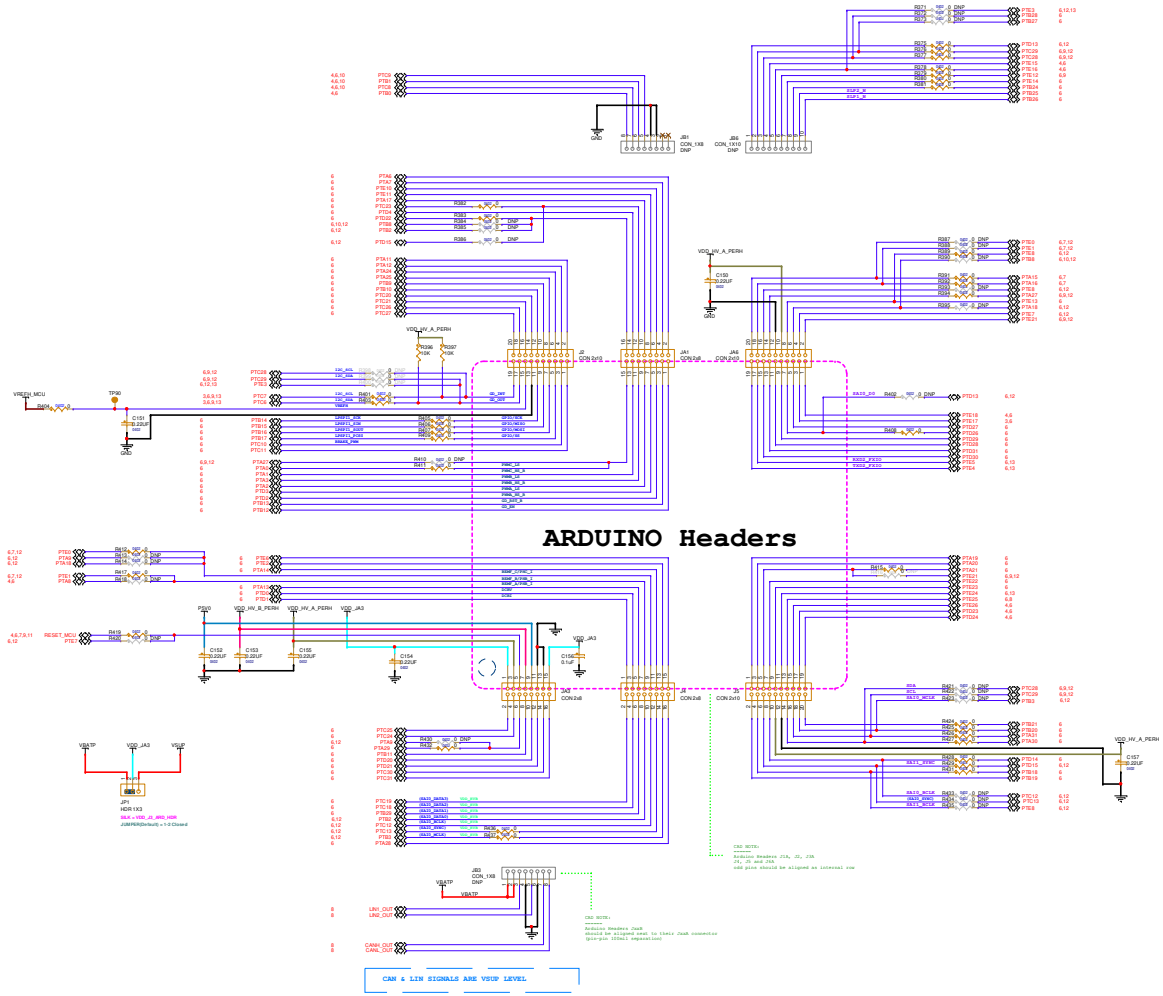
ADC Potentiometer



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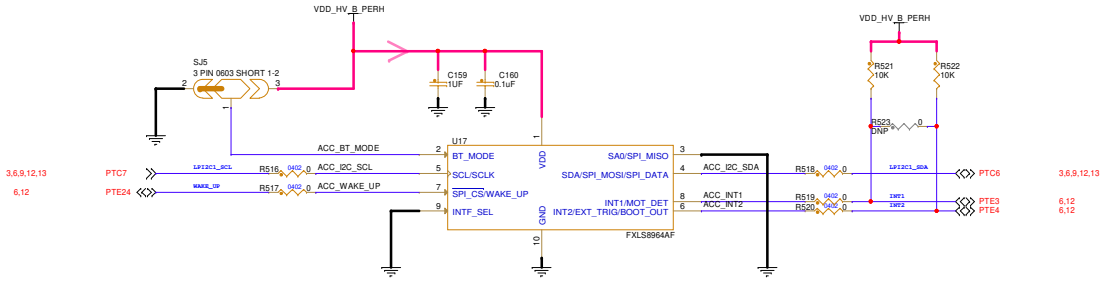
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Designer: Antonio Q. / Efren D.	Drawing Title: S32K344MINI-EVB
Drawn by: Antonio Quiroz	Page Title: QSPI Memory Interface
Approved: Efren Diaz	Size C / Document Number SCH-94921 PDF: SPF-94921 / Rev B1
	Date: Wednesday, October 01, 2025 / Sheet 11 of 14



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Designer: Antonio G. / Efren D.	Checked by: <CB>
Drawn by: Antonio Cuevas	Page Title: ARDUINO Shield
Approved: Efren Diaz	Size: D; Document Number: SCH-94921 PDF:SPF-94921; Rev: B1
	Date: Wednesday, October 01, 2025; Sheet 10 of 14

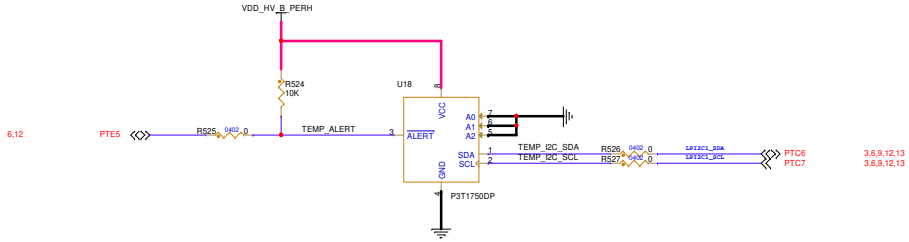
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ACCEL SENSOR



I2C ADDRESS: WR:30 RD:31

TEMP SENSOR



I2C ADDRESS: 0X90

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Designer: Antonio Q. / Efren D.	Drawing Title: S32K344MINI-EVB	
Drawn by: Antonio Quiroz	Page Title: SENSORS	
Approved: Efren Diaz	Size C / Document Number SCH-94921 PDF: SPF-94921	Rev B1
	Date: Wednesday, October 01, 2025	Sheet 13 of 14

JUMPER Table

REF DES	JUMPER(DEFAULT)	PAGE NAME
JP11	1-2 Closed	04. FS26
JP4,JP5	1-2 Closed	05. MCU PWR
J12	OPEN	07. OPENSDA K26
JP1	1-2 Closed	12. ARDUINO HDRS

DNP Table

REF DES	ASSY_OPT	PAGE NAME
R88,R123,R128,R448,R449,R455, R456,R457,R458,R502,R730, TPH23	DNP	03. USB TYPE-C/PD
C1,C9,D1,J10,R4,R42,R529, TPH2,TPH11	DNP	04. FS26
C62,C63,C64,Q5,R47,TPH3	DNP	05. MCU PWR
R50,R64,R65,R66,R67,R76,R79, R81	DNP	06. MCU PORTS1
C99,J12,R107,R115,R116,R117, R118,R138,R150,R153,TPH10, TPH16	DNP	07. OPENSDA K26
C104,C105,C113,C114,R167, R168,R170,R180,R181,R182, R454,R535,R536,TPH24	DNP	08. CAN/LIN PHY
C117,C118,C119,C120,R183, R184,R185,R186,R190,R191, R192,R194,R195,R196,R199, R200,R202,R206,R207,R208, R209,R212,R213,R214,R215, R221,R222,R225,R230,R231, R236,R237,R238,R239,R245, R246,R247,R248,R250,R251, R252,R253,R254,R255,R266, R268,R269,R305,R307,R308, R309,R311,R443,R537,R539, R542,R544	DNP	09. ETHERNET 10/100BASET
C133,C134,C135,C136,C137, C138,J8,R272,R273,R274,R275, R276,R277,R278,R279,R282, R283,R284,R285,R286,R287, R288,R289,R290,R291,R292, R293,R295,R296,R297,R298, R299,R300,R301,R302,R303, R304,TPH7,TPH12	DNP	10. ETHERNET CONNECTOR
R325,R326,R327,R328,R329, R330,R346,R350,R351,R352, R353,R368,TPH6,TPH19	DNP	11. QSPIA / JTAG / PERIPHERALS
JB1,JB3,JB6,R371,R372,R373, R384,R385,R386,R387,R388, R390,R393,R395,R398,R399, R400,R402,R410,R413,R414, R416,R418,R420,R421,R422, R423,R430,R433,R434,R435	DNP	12. ARDUINO HDRS
R523	DNP	13. SENSORS

SWITCH Table

REF DES	SWITCH(DEFAULT)	PAGE NAME
SW5	1-2 / 4-5	03. USB TYPE-C/PD

Shunt Table

REF DES	SHORT(DEFAULT)	PAGE NAME
SJ4	1-2	05. MCU PWR
SJ1,SJ2	1-2	07. OPENSDA K26
SJ5	1-2	13. SENSORS

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Designer: Antonio Q. / Efren D.	Drawing Title: S32K344MINI-EVB
Drawn by: Antonio Quiroz	Page Title: NOTES & BLOCK DIAGRAM
Approved: Efren Diaz	Size C; Document Number SCH-94921 PDF: SPF-94921; Rev B1
	Date: Wednesday, October 01, 2025; Sheet 14 of 14